Light

GCSE Computer Science – Network Protocols (Set 3)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Network Protocols

Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A tablet made in South Korea communicates successfully with a server located in Germany. What allows these devices to exchange data correctly?

- ○ **A.** Network protocols that define standardised rules for data transmission
- ○ **B.** Both devices using the same brand of processor
- ○ **C.** A direct cable connecting South Korea to Germany
- ○ **D.** Special translation software installed on each device

Q2 (1 mark)

A language learning app displays lesson content by connecting to web servers. Which protocol does the app use to request these web pages?

- ○ **A.** SMTP requests the lesson web pages
- ○ **B.** IMAP requests the lesson web pages
- ○ **C.** HTTP or HTTPS requests the lesson web pages
- ○ **D.** TCP alone requests the lesson web pages

Q3 (1 mark)

A food delivery app requires users to log in before ordering. Why should this login process use HTTPS rather than HTTP?

- ○ **A.** HTTPS displays the login form more attractively
- ○ **B.** HTTPS is required by law for all mobile apps
- ○ **C.** HTTPS allows users to log in faster
- ○ **D.** HTTPS encrypts login credentials during transmission, preventing interception

Q4 (1 mark)

A museum offers virtual tours where visitors view high-quality images of exhibits. Why is TCP's reliability important for this application?

- ○ **A.** TCP ensures all image data arrives complete, preventing corrupted or partial images
- ○ **B.** TCP makes the images appear in colour rather than black and white
- ○ **C.** TCP increases the resolution of the images during transmission

- ○ **D.** TCP stores backup copies of all images on the visitor's device

Q5 (1 mark)

Why does every device connected to the internet need a unique IP address?

- ○ **A.** To display the device owner's name on all websites visited
- ○ **B.** To measure how fast the device can download files
- ○ **C.** To identify the device so data packets can be correctly delivered to it
- ○ **D.** To determine which colour scheme the device should display

Q6 (1 mark)

An accountant emails a client's completed tax return. Which protocol transmits this email from the accountant's computer to the email server?

- ○ **A.** HTTP transmits the email to the server
- ○ **B.** IMAP transmits the email to the server
- ○ **C.** HTTPS transmits the email to the server
- ○ **D.** SMTP transmits the email to the server

Q7 (1 mark)

A sales representative checks work emails on both their laptop and phone, seeing identical inbox contents on each device. Which characteristic of IMAP enables this?

- ○ **A.** IMAP sends copies of each email to every device automatically
- ○ **B.** IMAP converts emails into a universal format for each device
- ○ **C.** IMAP requires users to forward emails to themselves
- ○ **D.** IMAP keeps emails stored on the server, allowing all devices to access the same messages

Q8 (1 mark)

A user browses a fashion retailer's website using a web browser. At which TCP/IP layer does the browser operate?

- ○ **A.** Link layer
- ○ **B.** Internet layer
- ○ **C.** Transport layer
- ○ **D.** Application layer

Q9 (1 mark)

TCP breaks data into packets and manages their reliable delivery. At which TCP/IP layer does TCP perform these functions?

- ○ **A.** Transport layer
- ○ **B.** Application layer
- ○ **C.** Internet layer
- ○ **D.** Link layer

Q10 (1 mark)

IP addresses data packets and determines their route across networks. At which TCP/IP layer does IP operate?

- ○ **A.** Application layer
- ○ **B.** Transport layer
- ○ **C.** Internet layer
- ○ **D.** Link layer

Q11 (1 mark)

What type of conversion happens at the link layer when data is transmitted across a network cable?

- ○ **A.** Web addresses are converted into IP addresses
- ○ **B.** Emails are converted into web pages
- ○ **C.** Digital data is converted into electrical signals for transmission
- ○ **D.** Images are converted into text descriptions

Q12 (1 mark)

When a car dealership in Manchester sends inventory data to headquarters in Birmingham, what does IP determine about the data packets?

- ○ **A.** The colour of cars in the inventory
- ○ **B.** The price of each vehicle in the database

- ○ **C.** The number of staff working at each location
- ○ **D.** The route packets take across networks between Manchester and Birmingham

Q13 (2 marks)

A florist's ordering system transmits customer orders to suppliers. Network congestion causes some packets to be lost. How does TCP ensure the complete order still arrives?

- ○ **A.** TCP abandons the transmission and displays an error message
- ○ **B.** TCP sends the entire order again from the beginning
- ○ **C.** TCP detects missing packets through sequence numbers and automatically retransmits only the lost packets
- ○ **D.** TCP converts the order to a smaller file format

Q14 (2 marks)

A sports league website sends live match results from servers in London to fans viewing on devices worldwide. What information does IP add to each data packet?

- ○ **A.** The source IP address (London server) and destination IP address (fan's device)
- ○ **B.** The final score of the match and player statistics
- ○ **C.** The fan's name and favourite team
- ○ **D.** The date and time the match was played

Q15 (2 marks)

A book publisher's website allows visitors to browse their catalogue. Describe how HTTP functions when a visitor clicks on a book to view its description.

- ○ **A.** HTTP automatically downloads all books to the visitor's device
- ○ **B.** HTTP encrypts the visitor's reading preferences
- ○ **C.** The browser sends an HTTP request to the server, which responds with the book description page
- ○ **D.** HTTP converts the book into an audio format

Q16 (2 marks)

A dental practice allows patients to view their treatment records online. Why is HTTPS essential for this patient portal?

- ○ **A.** HTTPS makes dental images display at higher resolution
- ○ **B.** HTTPS is only needed for financial websites, not medical ones
- ○ **C.** HTTPS speeds up access to patient records
- ○ **D.** HTTPS encrypts sensitive medical and personal information during transmission, preventing unauthorised access

Q17 (2 marks)

A concert venue emails tickets to customers who purchased them online. The customers access these tickets on multiple devices. Explain how email protocols are used.

- ○ **A.** IMAP sends the tickets; SMTP allows access on multiple devices

- ○ **B.** SMTP sends the ticket emails from the venue's server; IMAP allows customers to retrieve tickets on multiple devices because emails stay on the server
- ○ **C.** Only HTTP is used since tickets are purchased online
- ○ **D.** TCP handles all email functions without needing SMTP or IMAP

Q18 (2 marks)

What characteristic defines protocols that operate at the application layer of the TCP/IP model?

- ○ **A.** They convert data into physical signals for transmission
- ○ **B.** They assign IP addresses to all devices on the network
- ○ **C.** They directly support user-facing applications and services like web browsers and email programs
- ○ **D.** They break data into packets and ensure reliable delivery

Q19 (2 marks)

Before a weather app downloads forecast data, the transport layer establishes communication between the app and the weather server. What happens during this setup?

- ○ **A.** The app and server agree on settings like packet size for the data transfer
- ○ **B.** The weather forecast is immediately downloaded without any setup
- ○ **C.** Physical cables are connected between the app and server

- ○ **D.** IP addresses are assigned to both devices

Q20 (2 marks)

A real estate company's website sends property listings from headquarters to branch offices across the country. What does the internet layer do at each network point along the path?

- ○ **A.** It examines destination addresses and forwards packets toward the correct branch office
- ○ **B.** It displays the property listings for inspection at each point
- ○ **C.** It creates backup copies of all listings at each network location
- ○ **D.** It compresses the listing images to smaller sizes

Q21 (2 marks)

When a library's computer sends book reservation data across the network, what role does the network interface card (NIC) at the link layer perform?

- ○ **A.** The NIC searches the library database for available books
- ○ **B.** The NIC assigns IP addresses to the computer
- ○ **C.** The NIC provides physical connection to the network and converts data into signals for transmission
- ○ **D.** The NIC encrypts the reservation to prevent unauthorised access

Q22 (2 marks)

A veterinary supplier sends product catalogues electronically to clinics across Europe. Explain the different roles of TCP and IP in this transmission.

- ○ **A.** TCP and IP perform identical functions at the same layer
- ○ **B.** TCP ensures all catalogue data arrives complete and in order; IP addresses packets and routes them across European networks
- ○ **C.** IP breaks data into packets; TCP determines the route to each clinic
- ○ **D.** TCP encrypts the catalogues; IP compresses them

Q23 (2 marks)

A fitness app synchronises workout data between users' devices and company servers in multiple countries. What makes this global communication possible?

- ○ **A.** Special agreements between each pair of countries
- ○ **B.** Network protocols that provide internationally standardised rules followed by all devices regardless of location
- ○ **C.** Physical cables connecting every country directly
- ○ **D.** Separate versions of the app for each country

Q24 (3 marks)

A gaming tournament streams match footage from an arena to viewers worldwide. Describe how the video data flows through all four TCP/IP layers when being transmitted from the arena.

- ○ **A.** Link → Internet → Transport → Application: data starts at the cable and moves up to the streaming software

○ **B.** Application → Transport → Internet → Link: the streaming software prepares video, TCP segments it for delivery, IP adds addresses and routes, the NIC transmits signals

○ **C.** Transport → Application → Link → Internet: TCP must process data before the streaming software creates it

○ **D.** All four layers process the data simultaneously for maximum speed

Q25 (3 marks)

A university student downloads lecture recordings from the institution's servers. Describe how the recording data passes through the TCP/IP layers on the student's laptop when being received.

○ **A.** Application → Transport → Internet → Link: the media player sends the request first, so data flows downward

○ **B.** Link → Internet → Transport → Application: the NIC receives signals, IP verifies the destination, TCP reassembles data in order, the media player displays the recording

○ **C.** Transport → Link → Application → Internet: TCP processes everything before other layers are involved

○ **D.** Only the application layer is needed since the student is just watching videos

Q26 (3 marks)

An architectural firm sends complex CAD drawings to clients internationally. These technical files must arrive exactly as created. Explain all the mechanisms TCP uses to guarantee accurate delivery.

○ **A.** TCP only checks that files have the correct extension; content accuracy is not verified

○ **B.** TCP assigns sequence numbers for correct ordering, sends acknowledgments to confirm receipt, uses checksums to detect errors, and automatically retransmits any lost or corrupted packets

○ **C.** TCP relies on the recipient to manually check the drawings for errors

○ **D.** TCP sends files three times and deletes any copies that differ

Q27 (3 marks)

A printing company emails proofs to clients for approval and receives feedback replies. Staff access these conversations from office computers, home laptops, and mobile phones. Describe the complete role of email protocols.

○ **A.** A single protocol handles all email functions for simplicity

○ **B.** SMTP transmits outgoing proofs and feedback to email servers; IMAP retrieves messages and synchronises them across all devices because emails remain stored on the server

○ **C.** Each device type requires a different email protocol

○ **D.** HTTP handles email when accessed from web browsers, making SMTP and IMAP unnecessary

Q28 (3 marks)

An international news organisation distributes articles from journalists in Tokyo, Cairo, and São Paulo to readers worldwide. Explain how IP enables this global distribution.

○ **A.** IP requires dedicated cables between each city and every reader location

○ **B.** IP assigns unique addresses to all devices, adds source and destination addresses to packets, and routes articles through multiple

intermediate networks regardless of geographical distance

○ **C.** Different protocols are needed for each continent, with IP only working within single countries

○ **D.** IP broadcasts all articles to every internet-connected device simultaneously

Q29 (3 marks)

An employee benefits portal allows staff to view payslips, update pension contributions, and access health insurance details. Evaluate the protection HTTPS provides throughout a user's session.

○ **A.** HTTPS only protects the login page; subsequent pages use faster unencrypted HTTP

○ **B.** HTTPS encrypts all session traffic including login credentials, payslip data, pension information, and health details, preventing interception throughout the entire session

○ **C.** HTTPS is unnecessary because employee data is not as sensitive as banking information

○ **D.** HTTPS protection ends once the user is logged in

Q30 (3 marks)

A hotel chain's booking website allows guests to search rooms, make reservations with payment, and receive confirmation emails. Identify which protocols handle each component of this system.

○ **A.** HTTP handles everything since all functions are on the same website

○ **B.** SMTP handles room searches, TCP handles payments, IMAP sends confirmations

○ **C.** HTTPS handles the web interface for searching and secure payment processing; SMTP sends confirmation emails from the hotel's server to guests

○ **D.** Different protocols are only needed if the website has multiple pages

Total: **55** marks • Time: **30 min**

↑ Back to top

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